

Fabric.js Integration in canvas.tsx

A Comprehensive Technical Guide to the Layered Canvas Architecture

1. The Dual-Layer Canvas Architecture

Previously, canvas.tsx only displayed rendered PDF pages, while canvas2.tsx was an isolated white canvas. To allow drawing and editing directly on top of PDF documents, we merged them into a synchronized Dual-Layer Architecture:

- Layer 0 (Bottom): A standard HTML5 canvas that renders the PDF page bitmap via PDF.js.
- Layer 1 (Top): An interactive Fabric.js canvas positioned absolutely at (top: 0, left: 0) with a transparent background, allowing the PDF document underneath to remain fully visible.

[Document Viewport Container] (relative, shadow-2xl, border)

```
| -- Layer 0: <canvas ref={pdfCanvasRef} className='block' /> (PDF Bitmap)

| -- Layer 1: <div className='absolute top-0 left-0 z-10'><canvas ref={fabricCanvasElRef} /></div>
              (Fabric.js Transparent Interactive Annotation Overlay)
```

2. Key State and References Introduced

- **canvasSize** ({ width, height }) : Stores current pixel dimensions of the PDF page.
- **pdfCanvasRef** (HTMLCanvasElement) : Direct reference to the bottom PDF bitmap canvas.
- **fabricCanvasElRef** (HTMLCanvasElement) : Direct reference to the upper Fabric canvas element.
- **fabricInstanceRef** (FabricCanvas) : Active instance of Fabric Canvas for imperative operations.
- **fabricCanvas** (FabricCanvas (State)) : Allows the Settings component to trigger reactive UI re-renders.
- **pageAnnotationsRef** (Record<number, any>) : Dictionary holding serialized JSON of shapes per page.

3. Zoom Synchronization & Mathematical Precision

Problem: When scaling a PDF document using PDF.js, only the PDF background was resizing. Fabric shapes stayed the exact same pixel size and drifted out of position relative to the document.

Solution: Fabric.js native `setZoom(scale)` method:

In PDF.js, document coordinates are calculated in unscaled typographical points (72 pt = 1 inch):

$$\text{viewport_pixel} = \text{pdf_point} * \text{scale}$$

By setting Fabric internal zoom equal to the PDF scale factor (`fabricCanvas.setZoom(scale)`),

Fabric coordinates become 1:1 with PDF points:

$$\text{fabric_pixel} = \text{fabric_point} * \text{scale}$$

Result: Every drawn rectangle, circle, and text element stays permanently glued to its exact spot on the PDF document at 50%, 100%, 200%, or any custom zoom level.

```
// Synchronizing Fabric Dimensions & Zoom upon PDF page render:
const viewport = page.getViewport({ scale });
pdfCanvas.width = viewport.width;
pdfCanvas.height = viewport.height;
setCanvasSize({ width: viewport.width, height: viewport.height });

if (fabricInstanceRef.current) {
  fabricInstanceRef.current.setDimensions({ width: viewport.width, height: viewport.height });
  fabricInstanceRef.current.setZoom(scale); // Scales all elements in sync with PDF
  fabricInstanceRef.current.renderAll();
}
```

4. Per-Page Annotation Persistence

In multi-page documents, annotations created on Page 1 must not appear on Page 2.

We implemented page-level serialization and restoration:

1. When `pageNumber` changes:

- Serialize current shapes to JSON: `pageAnnotationsRef.current[prevPage] = fCanvas.toJSON()`
- Clear the canvas: `fCanvas.clear()`
- If annotations exist for the new page: `fCanvas.loadFromJSON(savedData)`
- Immediately re-apply `fCanvas.setZoom(scale)` to preserve the active zoom level.

5. Integrated Toolbars & Shape Creation Handlers

- **addRectangle()** : Creates a `Rect({ width: 120, height: 80, fill: '#3b82f6' })` and sets it active.
- **addCircle()** : Creates a `FabricCircle({ radius: 50, fill: '#ef4444' })` and sets it active.
- **addText()** : Creates an editable `Textbox({ fontSize: 22, fill: '#1f2937' })` and sets it active.

6. Properties Panel & UI Layout

- **Left Floating Palette:** Fixed pill on the left (Square, Circle, Type icons) to add shapes anytime.
- **Right Properties Panel:** Mounts `<Settings canvas={fabricCanvas} />`. Selecting any shape opens controls for width, height, circle diameter, text font size, and color picker.
- **Bottom-Right Zoom Bar:** Zoom Out, Percentage display (click to reset), Zoom In, and Reset buttons.

Architecture Verification & Quality Checklist

- [x] Dual-Layer Alignment: PDF canvas and Fabric overlay are pixel-aligned at (0, 0).
- [x] Synchronized Zoom: `fabricCanvas.setZoom(scale)` guarantees 1:1 vector scaling.
- [x] Zero TypeScript Errors: Passes `npx tsc --noEmit` with 0 issues.
- [x] Dynamic Cleanup: All render tasks and Fabric instances are safely disposed on unmount.